NAKIB AHMED RAJ

M.Sc. / Ph.D. Applicant · Artificial Intelligence & Natural Language Processing
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RESEARCH INTERESTS

My primary research interests lie at the intersection of Natural Language Processing (NLP), Trustworthy AI, and Scalable Machine Learning Systems and AI Agents. Specific areas of focus include:

- Multi-task and transfer learning for low-resource NLP tasks, particularly hate speech, sentiment, and aggression detection in multilingual social media contexts
- Robust and interpretable deep learning architectures for text classification, incorporating Transformer-based models (BERT, RoBERTa), convolutional-recurrent hybrids (CNN-BiLSTM), and multi-head self-attention mechanisms
- MLOps and production-grade machine learning systems: pipeline reproducibility, model drift detection, and large-scale data engineering
- Fairness, bias mitigation, and generalization in neural language models
- Agentic AI and LLM-based systems, focusing on tool orchestration, agent state and context management, execution reliability, and developer-centric agentic workflows

EDUCATION

B.Sc. in Mechatronics and Industrial Engineering *2020 – 2025*
Chittagong University of Engineering and Technology (CUET), Chittagong, Bangladesh

- CGPA: 3.27 / 4.00
- Undergraduate Thesis: "Decoding the Spectrum of Online Harassment: A Neural Approach to Sentiment, Hate and Aggression Identification"
- Relevant Coursework: Artificial Intelligence, Digital Signal Processing, Control Systems, Numerical Methods, Engineering Mathematics

Higher Secondary Certificate (HSC) *2017 – 2019*
Mirpur Cantonment Public School and College, Dhaka

- GPA: 5.00 / 5.00 (Science)

Secondary School Certificate (SSC) *2015 – 2017*
Adamjee Cantonment Public School, Dhaka

- GPA: 5.00 / 5.00

RESEARCH EXPERIENCE

Undergraduate Thesis Research *2024 – 2025*
Department of Mechatronics and Industrial Engineering, CUET
Thesis: *"Decoding the Spectrum of Online Harassment: A Neural Approach to Sentiment, Hate and Aggression Identification"*
Overview:
Conducted a systematic investigation into automated online harassment detection using multi-task deep learning. The work addresses three inter-related classification problems like emotion/sentiment analysis, violence/aggression detection, and hate speech classification within a unified neural framework applied to social media text.
Architectures & Methodologies:

- Designed and implemented a hybrid neural encoder combining a from-scratch Transformer, a multi-scale

- CNN, and a stacked BiLSTM with self-attention pooling, fused with hand-crafted linguistic features (length, polarity, word/char counts) for joint emotion, violence, and hate-speech classification
- Benchmarked the proposed model against five baselines, BERT-base, RoBERTa-base, DeBERTa-base, LSTM-only, and CNN-only under an identical multi-task head configuration
- Designed a multi-task learning (MTL) framework sharing lower encoder layers across all three classification tasks, with task-prefix routing 'EMOTION', 'VIOLENCE', 'HATE' and task-specific residual adapter heads, enabling transfer of cross-task linguistic features and reducing task-specific overfitting
- Built a reproducible end-to-end training pipeline using AdamW with cosine warmup, mixed-precision training, gradient checkpointing, multi-GPU via DataParallel, weighted cross-entropy, and task-balanced sampling
- In progress: attention visualisation (attention rollout) to interpret model decision boundaries and audit for demographic bias

Datasets & Scale:

- Curated and pre-processed balanced corpora across three annotation schemes from public benchmarks, harmonised to approximately 150,000 labelled instances (50K per task) using a custom cleaning pipeline (contraction expansion, emoji demojizing, URL/mention/hashtag removal, concatenated-word splitting, NLTK tokenisation, stopword removal, lemmatisation) parallelised with joblib
- Applied manual class balancing via token-swap/token-drop augmentation, plus in-training synonym replacement (via nlpaug) and mixup interpolation between same-task sample pairs to address class imbalance

Evaluation & Results:

- Evaluated on a held-out 10% stratified test split using Accuracy, Macro-F1, Precision, and Recall per task; calibration quantified via ECE, Brier score, and reliability diagrams
- On the preliminary single-seed run (seed 42): Emotion Accuracy 0.957 / Macro-F1 0.957; Violence Accuracy 0.995 / Macro-F1 0.995; Hate speech Accuracy 0.969 / Macro-F1 0.968, with 224.7M parameters, 5.7 GB peak VRAM, and ~22 s inference on the test split (T4 GPUs)
- In progress: stratified k-fold cross-validation (k=5), Matthew's Correlation Coefficient (MCC) reporting, multi-seed statistical significance (5–10 seeds with mean ± std, 95% CI, and p-values vs strongest baseline), and learning-curve analysis across 20/40/60/80/100% training-data fractions
- In progress: robustness analyses including out-of-domain evaluation on an unseen external hate-speech corpus and adversarial text perturbation under 7 attack types (character substitution, misspellings, word deletion, synonym, punctuation, emoji insertion, slang/obfuscation) to demonstrate model generalisation across writing styles and platform dialects

Research Contributions:

- Proposed a unified annotation taxonomy reconciling disparate labelling conventions across existing hate speech corpora
- Investigated intersectional bias in pre-trained language models by probing representations for protected attributes; findings inform fairness-aware fine-tuning procedures
- Manuscript currently in preparation for submission to an NLP/AI venue (venue TBD)

INDUSTRY & PROFESSIONAL EXPERIENCE

Junior Software Engineer – MLOps, Data Engineering, AI Engineering *January 2025 – Present*
Poridhi.io, Dhaka, Bangladesh

- Architected and deployed scalable real-time and batch data processing pipelines using Apache Spark (distributed DataFrame operations over multi-node clusters), Apache Kafka (event streaming at >50k messages/sec throughput), and Apache Flink (stateful stream processing with exactly-once semantics)
- Engineered production-grade ML systems for churn prediction (gradient-boosted and neural classifier ensembles) and fraud detection (anomaly detection via Isolation Forest + LSTM sequence models), covering the full lifecycle from raw data ingestion to containerised model serving
- Implemented comprehensive model observability frameworks integrating Prometheus (metric scraping), Grafana (real-time dashboards), and EvidentlyAI (statistical drift detection on feature and prediction distributions), reducing undetected model degradation incidents

- Provisioned and managed cloud infrastructure on AWS (EC2, S3, RDS, ECR) using Pulumi Infrastructure-as-Code, ensuring reproducible, version-controlled deployments
- Managed end-to-end experiment tracking, model versioning, and staged promotion using MLflow Model Registry, enabling systematic A/B comparison across >30 experimental runs

Software Engineering Intern – Data & ML Infrastructure *October 2024 – December 2024*
Poridhi.io, Dhaka, Bangladesh

- Designed and maintained automated ETL and ML data preparation workflows using Apache Airflow DAGs, reducing manual pipeline intervention time by ~40%
- Containerised ML training and inference services with Docker, establishing reproducible environments across development, staging, and production
- Developed RESTful model inference APIs using FastAPI, implementing asynchronous request handling and input validation schemas
- Operated within Linux-based CI/CD workflows (GitHub Actions), maintaining code quality through pull-request reviews and automated testing

Industrial Attachment – BITAC, Dhaka
Bangladesh Industrial Technical Assistance Centre

- Completed structured training in industrial engineering processes, manufacturing systems, and automation integration
- Gained practical exposure to electro-mechanical systems and CNC-based manufacturing workflows

SELECTED TECHNICAL PROJECTS

Puku-CLI - Agentic Terminal Harness Engine (Active) *2025-2026*

- Designing and testing a CLI-native agentic framework built around hooks, skills, agents, and sub-agents, with a PUKU.md configuration spec (analogous to CLAUDE.md) governing agent behaviour, MCP integrations, and slash command dispatch
- Work explores harness architecture for composable, inspectable agentic pipelines, focusing on how structured context, tool routing, and inter-agent communication can be made reliable and debuggable at the terminal level

Multi-Task NLP Pipeline for Online Harassment Detection *2024 – 2025*

- End-to-end Python pipeline using HuggingFace Transformers, PyTorch, and Scikit-Learn for data preprocessing, model training, hyperparameter optimisation (Optuna), and evaluation reporting
- Trained on ~80,000 labelled social media instances; achieved macro-F1 of ~0.87 (hate) and ~0.83 (aggression) with fine-tuned BERT-base-uncased
- Experiment tracking via MLflow; results reproducible via Docker containerisation

Production Churn & Fraud Detection ML System *2025*

- Designed a dual-model production system deployed on AWS EC2, serving predictions via FastAPI endpoints with sub-200ms P95 latency
- Integrated EvidentlyAI drift reports and Grafana alerting for continuous monitoring; automated retraining triggers on PSI threshold breach

Real-Time Streaming Analytics Platform *2025*

- Built a Kafka → Flink → ClickHouse pipeline for real-time event analytics, processing >50,000 events/second with windowed aggregation and fault-tolerant checkpointing
- Provisioned infrastructure using Pulumi on AWS; end-to-end latency under 500ms for live dashboard updates

LLM-Powered Retrieval-Augmented Generation (RAG) Prototype *2024*

- Developed a document Q&A system using LangChain and LangGraph orchestration, Qdrant vector database for semantic retrieval, and a locally hosted LLM backend

- Instrumented with Langfuse for LLM observability, tracking latency, token usage, and retrieval quality metrics

PUBLICATIONS & PRESENTATIONS

Manuscripts in Preparation:

- Raj, N. A. (in preparation). "Decoding the Spectrum of Online Harassment: A Multi-Task Neural Approach to Sentiment, Hate and Aggression Identification." Target venue: ACL/EMNLP/COLING (NLP track). Expected submission: 2026.

Conference Participation:

- Active member of IEEE Computer Society Student Branch Chapter, CUET – Machine Learning Wing; participated in internal seminars and technical workshops on AI and data systems.

TECHNICAL SKILLS & COMPETENCIES

Programming & Query Languages:
Python (primary), SQL

Machine Learning & Deep Learning Frameworks:
PyTorch, HuggingFace Transformers, Scikit-Learn, NumPy, Pandas, OpenCV, LangChain, LangGraph

Distributed & Scalable Data Systems:
Apache Spark, Apache Kafka, Apache Flink, Apache Airflow

MLOps & Experiment Management:
MLflow, EvidentlyAI, Prometheus, Grafana, Langfuse

Cloud Platforms & Infrastructure-as-Code:
Amazon Web Services (EC2, S3, ECR, Lambda), Docker, Pulumi, Terraform

Databases & Vector Stores:
PostgreSQL, MongoDB, ClickHouse, Qdrant

API & Service Development:
FastAPI, RESTful API design, asynchronous Python (asyncio)

Development & Reproducibility:
Git, GitHub Actions (CI/CD), Linux (Ubuntu/Debian), Docker Compose

LEADERSHIP & PROFESSIONAL SERVICE

Secretary, Machine Learning Wing *CUET, 2023 – 2025*
IEEE Computer Society Student Branch Chapter, Chittagong University of Engineering and Technology

- Organised technical workshops and study groups on ML fundamentals, deep learning, and NLP for undergraduate members
- Coordinated inter-university academic activities and maintained correspondence with IEEE regional bodies

REFERENCES

Available upon request. Academic and professional referees can be provided from CUET (thesis supervisor) and Poridhi.io (industry mentor).